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Evaluating the impact of a synthetic predator pheromone, a novel pest management tool, on target pests and insect communities in potato fields

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Abstract:

Despite their reputation for species specificity, pheromones can influence interactions across trophic levels. This is particularly relevant in agricultural systems where predator-derived cues could influence pest and natural enemies. The predatory spined soldier bug, *Podisus maculiventris* (Say) (Hemiptera: Pentatomidae), preys upon the Colorado potato beetle, *Leptinotarsa decemlineata* (Say) (Coleoptera: Chrysomelidae), a major pest of solanaceous crops. Previous field work shows that a synthetic aggregation pheromone of *P. maculiventris* reduces beetle larval abundance through non-consumptive effects, yet potential impacts on other herbivores and predators remain unclear. We hypothesized that synthetic predator pheromone cues could alter herbivore and predator feeding behavior in an agroecosystem. A field experiment evaluated the effects of the synthetic predator pheromone on herbivore and natural enemy abundance and diversity. We also deployed sentinel *L. decemlineata* egg clutches across the season to measure field predation rates. Greenhouse feeding assays tested whether common predators altered consumption of *L. decemlineata* eggs in response to the pheromone. Field results indicated a reduction in *L. decemlineata* larval abundance in pheromone-treated plots without detectable changes in overall herbivore or predator community abundance or diversity. However, predation rates in pheromone-treated plots were five times higher compared to the control. Greenhouse assays showed increased egg consumption by *P. maculiventris*. These findings suggest that the synthetic predator pheromone may reduce *L. decemlineata* populations by enhancing foraging behavior of *P. maculiventris* providing a potential tool for potato pest management with minimal non-target impacts on herbivore and predator communities.

Keywords: chemical ecology, predator cues, non-consumptive effects, foraging behavior, behavioral control

Introduction

Semiochemicals, especially sex pheromones, have become essential tools in agricultural pest management, with applications ranging from population monitoring and mass trapping to attract-and-kill, push–pull systems, and mating disruption. Their high specificity and relatively low environmental footprint make them attractive components of integrated pest management (IPM) programs (Sharma et al. 2019, Rizvi et al. 2021). For example, sex pheromones are widely deployed to disrupt mating in orchard pests such as the codling moth (*Cydia pomonella*), where pheromone-based disruption has reduced insecticide inputs and becomes a cornerstone of apple and pear IPM (Light et al. 2001, Witzgall et al. 2008). Along with sex pheromones, aggregation pheromones represent a promising new approach to manipulate pest populations because they attract and retain both sexes. For instance, an aggregation pheromone blend common across thrips has been used to manage greenhouse and field pests, including western flower thrips (*Frankliniella occidentalis*), through attract-and-kill strategies and behavioral interference (Kim et al. 2023). Another notable example that works in field settings is vittatalactone, an aggregation pheromone that successfully attracts both male and female striped cucumber beetles (*Acalymma vittatum*)(Weber et al. 2023).

However, the long-held assumption that pheromones act with strict species specificity is increasingly challenged with abundant evidence that pheromones influence interactions across multiple trophic levels. For instance, the bark beetle aggregation pheromone, ipsdienol, acts as an intraspecific attractant but also as a kairomone for its predator, the checkered beetle (*Thanasimus dubius*) demonstrating how pheromone signals can be co-opted across trophic

levels (Dahlsten et al. 2004). In addition, unlike sex pheromones, many aggregation pheromones are multicomponent blends produced and used by multiple taxa. These blends often include plant-derived volatiles or structurally similar convergent compounds, making them more likely to influence non-target organisms. For example, in the squash (*Cucurbita pepo*) system, squash bug (*Anasa tristis*) has been shown to eavesdrop on vittatalactone, the male-produced aggregation pheromone of the co-occurring striped cucumber beetle (*A. vittatum*) and increase its oviposition (Brzozowski et al. 2022). In another example, non-host green leaf volatiles (GLVs) from deciduous trees can also inhibit bark beetle pheromone responses, discouraging colonization of unsuitable hosts (Huber et al. 2021). Collectively, these examples demonstrate that semiochemicals often function through the community, influencing herbivores, predators, and parasitoids. As agroecosystems contain multiple interacting taxa, assessing these cross-level effects is essential for the ecologically safe and effective deployment of pheromone-based tools.

Pheromones and predator-associated volatiles can elicit strong non-consumptive effects (NCEs) on prey as individuals alter behavior to avoid predation. Exposure to predator odors can trigger dispersal, reduce feeding, and deter oviposition, ultimately suppressing pest populations. For example, aphids detect and avoid surfaces contaminated with chemical cues from lady beetles (Ninkovic et al. 2013) and reduce phloem ingestion in the presence of a lady beetle volatile cue (Kansman et al. 2023). Such behavioral shifts represent an underexplored yet promising avenue for pest suppression, where volatiles can deter pests and suppress feeding, leading to yield improvements (Martinez et al. 2025). However, as with other semiochemicals, predator cues may also affect non-target species. Predator-derived volatiles are likely to influence a wide array of arthropods, including beneficial predators and alternative prey, thereby reshaping community interactions. For example, Wen et al. (2023) found that a rove beetle

pheromone spray reduced predatory spiders while overall arthropod diversity remained stable, indicating limited off-target effects. Still, many predatory insects respond to volatile cues that overlap chemically with herbivore-induced plant volatiles (HIPVs), which are broadly detectable by multiple organisms and can generate non-target and community-level effects (Kaplan 2012).

This study focuses on one such system involving the spined soldier bug, *Podisus maculiventris* (Say) (Hemiptera: Pentatomidae), a generalist predator and key natural enemy of the Colorado potato beetle (*Leptinotarsa decemlineata*)(Say) (Coleoptera: Chrysomelidae). Male *P. maculiventris* produces an aggregation pheromone that attracts both sexes and can modify prey responses, including reducing leaf consumption and larval abundance (Hermann and Thaler 2014, Aflitto and Thaler 2020). There are several hypotheses for why this pheromone may affect the community of herbivores or natural enemies in the system. Because this cue aggregates conspecifics, it may intensify intraspecific competition among naturally occurring or augmented *P. maculiventris* when prey are limited or alter intraguild interactions. *P. maculiventris* is known to act as an intraguild predator on the juveniles of other soft-bodied predators like lady beetles (Hough-Goldstein et al. 1996, Mallampalli et al. 2002). Conversely, herbivores may interpret the same cue as evidence of increased predation risk and adjust their behavior by reducing feeding, avoiding treated areas, or altering oviposition to lower the likelihood of predator encounter (Werner and Peacor 2003, Hermann and Thaler 2014). The synthetic pheromone blend consists of α -terpineol, terpinen-4-ol, benzyl alcohol, linalool, and (*E*)-2-hexenal (Aldrich et al. 1984). Two components, linalool and (*E*)-2-hexenal, are also green leaf volatiles released by potato plants following herbivore injury, and herbivore-induced plant volatiles (HIPVs) are broadly used by insects across taxa as indicators of herbivore activity, plant damage, and changes in the local risk landscape (Heil 2014). This chemical overlap between predator cues and plant volatiles

suggests potential for community-level, non-target effects when the synthetic blend is deployed in the field.

In this study, we evaluate how a synthetic blend mimicking *P. maculiventris* aggregation pheromone influences the target pest the Colorado potato beetle and the associated arthropod communities as well as enemy-pest interactions in potato fields. We hypothesize that herbivores and predators may alter their feeding behavior in response to *Podisus*-specific cues, pheromone-induced changes in the abundance or feeding behavior of naturally occurring predators. 1) We used field experiments to test the effects of the synthetic predator pheromone on herbivore and natural enemy abundance and diversity using weekly D-vac sampling and visual counts. 2) We tested the effects of pheromone on predation rates by deploying sentinel beetle egg clutches across the potato growing season. 3) We conducted complementary greenhouse assays to test whether the pheromone affected predation rates of beetle eggs by dominant field predators. Understanding these dynamics is crucial for assessing the non-target and community-level impacts of deploying predator pheromones as a novel biocontrol tool in agroecosystems.

Methods

1. Field experiments

Field set-up

On May 3, 2024, *Solanum tuberosum* (cv. Yukon Gold) potatoes were planted at Cornell University's Homer C. Thompson Vegetable Research Farm in Freeville, NY (42.517895, -76.332011). The variety Yukon Gold was selected due to its extensive use in prior research in this system (Hermann and Thaler, 2014; Aflitto and Thaler, 2020). Before planting, the field was fertilized with Triple 13 NPK at a rate of 1289 kg/ha. The experimental layout consisted of 36

single rows of potatoes, each row measuring 30 meters in length. Potato seeds were planted 0.3 m apart within rows, with a row spacing of 0.9 m. The field was organized into three blocks (A, B, and C), each containing 12 parallel rows. Blocks were separated by approximately four meters, oriented from right (A) to center (B) to left (C) of the site.

Predator pheromone release

The synthetic predator pheromone used in this study mimics the male aggregation pheromone of *P. maculiventris*, which is naturally released in spring to attract females. The dispenser containing the predator pheromone was made from a modified opaque brown 2 mL Eppendorf tube with three 0.55 mm holes in the top. Each dispenser was filled with 594.1 μ L of a synthetic male *P. maculiventris* dorsal abdominal gland pheromone mixture (354.0 μ L (E)-2-hexenal, 216.0 μ L alpha terpineol, 2.4 μ L linalool, 2.0 μ L terpinen-4-ol, and 19.7 μ L benzyl alcohol) (Aldrich et al., 1984) and affixed to a highlighter green marking flag 20 cm high. Control plots received an empty tube attached to a marking flag. Pheromone dispensers have been shown to exert non-consumptive effects within approximately one meter of their placement (Martinez et al. 2025; Aflitto and Thaler 2020). To account for this, plots were designed to be two meters long and included two pheromone dispensers per plot. The dispensers emitted the pheromone at an average rate of 0.23 mg/hr (Aflitto and Thaler 2020). Studies have found that a single male *P. maculiventris* can release up to a maximum of 1 mg of the pheromone at a time, indicating that this blend and release rate mimics what the Colorado potato beetle could potentially encounter in the field (Aflitto & Thaler, 2020; Aldrich et al., 1984).

Pheromone treatments were deployed on May 30, 2024 to coincide with the emergence of overwintering Colorado potato beetle adults. Within each row, plots consisted of six plants spaced 30 cm apart in a single row, with two pheromone flags placed adjacent to the second and

fourth plants. A five meter buffer of untreated potato plants was maintained between plots. To prevent side-by-side overlap of treatments, plots were staggered between rows.

Each row contained two pheromone-treated plots and two control plots. Pheromone-treated plots received two open-dispersion volatile dispensers, while control plots received two empty dispensers. Each plot type included two distinct sampling methods, as described below, and treatments were randomized across all rows for a total of 144 plots. Dispensers were left in the field throughout the measurement period and were replaced every 1.5 weeks.

Field sampling

We measured the effect of predator pheromone on the abundance of potato herbivores and natural enemies using two sampling methods. Starting June 4, 2024, and continuing weekly for seven weeks, we conducted visual counts of all Colorado potato beetle predators, the Colorado potato beetle (eggs, larvae, and adults), and other potato pests present on the six plants within each of the 72 plots (half control, half with pheromone). For each plant, the observers positioned themselves approximately 15–30 cm from the plant material and conducted a systematic inspection. Observers gently manipulated the stems and foliage to examine both the upper and lower surfaces of all leaves, including those near the base of the plant, to ensure full coverage of the plant canopy. Insects were identified visually in the field and were not collected during visual surveys.

In addition to weekly visual estimates, we used a handheld D-vac device (Model 122 Rincon-Vitova Insectaries, Ventura CA) on the other 72 plots (half control, half with pheromone). Each plant was vacuumed for 5–10 seconds. Samples from all six plants were placed into a collection bag labeled with block, plot ID, row, date of collection and placed in the freezer.

Insect Identification

All observed organisms were identified to family, with taxa of particular importance for Colorado potato beetle predation and potato pest status identified to finer taxonomic resolution when possible. Focal herbivores included the potato leafhopper, *Empoasca fabae* (Harris) (Hemiptera: Cicadellidae); the Colorado potato beetle, *Leptinotarsa decemlineata* (Say) (Coleoptera: Chrysomelidae); aphids (Hemiptera: Aphididae); the tarnished plant bug, *Lygus lineolaris* (Palisot de Beauvois) (Hemiptera: Miridae); and flea beetles (Coleoptera: Chrysomelidae: Galerucinae) (Weber et al. 2022). Predators were identified to family and included lady beetles (Coleoptera: Coccinellidae), minute pirate bugs (Hemiptera: Anthocoridae), spiders (Araneae: Araneidae), damsel bugs (Hemiptera: Nabidae), and lacewings (Neuroptera: Chrysopidae, Hemerobiidae). Species-level identification was conducted for key predators of the Colorado potato beetle, including *Lebia grandis* (Coleoptera: Carabidae) (Hentz) and *Podisus maculiventris* (Hemiptera: Pentatomidae) (Say). Lady beetles were further identified as *Coleomegilla maculata* (Coleoptera: Coccinellidae) (De Greer), a native, naturally occurring predator of Colorado potato beetle. Herbivore and predator taxa representing less than 1% of total observations were not identified further, as these uncommon taxa were unlikely to play a substantial role in pest suppression or plant damage within this system (Chang and Snyder 2004).

Egg predation rates in the field

To quantify Colorado potato beetle egg predation rates in the field over the potato growing season, we deployed sentinel beetle eggs weekly from June 7 to July 19, 2024. We prepared sentinel eggs using three-week-old potted potato plants grown in the greenhouse and fresh egg clutches (20 to 40 eggs per clutch) collected from a lab colony of the Colorado potato beetle. We used a paired design where an egg clutch was glued to the lower side of a leaf (the “predator-exposed” clutch)

and another clutch was glued to the lower side of a nearby leaf of the same plant enclosed in a mesh bag (the “control” clutch). Ten to fourteen plants with sentinel eggs were deployed in the field for three days each week (giving a total sample size of 84 over the season), with half of the plants in the control plots and half of the plants in the pheromone-treated plots (the plants with sentinel eggs were placed between two field potato plants). The number of lost eggs in each clutch was determined as the difference between the initial number of eggs and the number of remaining intact eggs (partially damaged and shrunk eggs were counted as lost eggs as they were not able to hatch). To account for non-predation egg mortality, background egg loss from the paired control clutch was scaled to the size of the predator-exposed clutch and subtracted from observed egg loss to estimate predation; clutches that were missing or hatched were excluded.

2. Lab experiments

Predation assays

Greenhouse feeding assays were conducted using the dominant Colorado potato beetle egg predators that were also abundant in the field, focusing on the species that could be reliably collected or maintained under laboratory conditions. These included *Coleomegilla maculata* adults and fourth-instar larvae, *Lebia grandis* adults, *Chrysoperla rufilabris* third-instar larvae, and *P. maculiventris* adults. The assays took place in the greenhouse facility at Cornell University in Ithaca, NY. Minute pirate bug *Orius insidiosus* was also assessed through this study due to its high abundance in the field, however there was no clear consumption evidence of eggs, and no further analysis was performed. *Coleomegilla maculata* adults, *Lebia grandis* and *P. maculiventris* adults were field collected from experimental plots located in Freeville, NY in August 2024. *Chrysoperla rufilabris* and *Orius insidiosus* were purchased from ARBICO Organics, Oro Valley, AZ. *Coleomegilla maculata* adults and fourth-instar larvae were taken

from our lab-colony reared in a growth chamber maintained on a 16:8 L:D dark cycle and fed on pea aphids.

For the greenhouse assays, predators were randomly selected from the lab colony once predators started reproducing or assays were done following field collection. Prior to the experiment, predators were isolated individually without food for 24 hours to standardize hunger levels.

Each replicate contained a single potato plant with a set number of *Leptinotarsa decemlineata* eggs affixed to the abaxial side of a single, mid-height leaf blade. The beetle eggs were collected from lab-maintained colonies in a growth chamber on a 16:8 L:D dark cycle. Temperature was maintained at approximately 28.2°C and a relative humidity of 21.4%. All eggs were collected within 24 hours before the start of the assays. Plants were spaced equally in the greenhouse, about 0.5 meters apart. Individual predators were introduced to the plants randomly assigned to either the control or the treatment group. Plants in the control group were provided rubber septas charged with 250 microliters of hexane solution and soaked overnight. Plants in the treatment group had rubber septas charged with 250 microliters of 1% pheromone solution (see above) diluted with hexane, attached to a leaf around 10-15 cm across from the egg-affixed leaf. All plants were covered by perforated bread bags and secured at the base of the plant with a rubber band. Replicates were evenly divided between control and pheromone treatments. Due to space constraints, the experiments were carried out over sequential trials, each containing an equal number of control and pheromone replicates. Total eggs consumed were counted after 24 or 48 hours. Predators that died or pupated during the assay were omitted from analysis.

The predation assays were done with *C. maculata* adults (n = 128; 63 control, 65 pheromone), *C. maculata* fourth-instar larvae (n = 27; 15 control and 12 pheromone), *L. grandis* adults (n = 32; 16 control and 16 pheromone), *C. rufilabris* third-instar larvae (n = 46; 23 control and 23

pheromone), and *P. maculiventris* adults (n = 62; 31 control and 31 pheromone; this assay was conducted for 48 hours).

3. Data analysis

Abundance of the Colorado potato beetle

We analyzed the abundance of the Colorado potato beetle (*L. decemlineata*) separately for the following biological and statistical reasons: (1) It is the focal pest species in this system. (2) It has known behavioral responses to predator-derived cues (Aflitto & Thaler, 2020; Helms et al., 2019). (3) It was not found in D-vac samples. (4) Its statistical distribution differed from that of other herbivores in the system. We fit a generalized linear mixed-effects model with beetle larval abundance as the response variable, pheromone treatment as the fixed effect, and week and plot nested within the block as the random effects.

Abundance, richness, and diversity of arthropod communities

To evaluate how pheromone treatment influenced herbivore and predator community structure, we calculated total abundance, mean abundance per taxonomic group, richness, and Shannon diversity for each field sample. Herbivore and predator datasets were first filtered to include only non-predatory taxa or predatory taxa only, and total counts were summed within each combination of block, row, plot, treatment, sampling method, week, and taxonomic group. Community metrics were calculated at the level of taxonomic groups (family-level units), including taxonomic richness (number of families detected), mean abundance per family, and Shannon diversity using the “vegan” package (Oksanen, 2024).

Abundance was log-transformed to meet model assumptions and analyzed with a linear mixed-effects model including treatment as the fixed effect and week and plot nested within block as

the random effects. Richness was modeled using a generalized linear mixed model with a Poisson error distribution, with treatment included as the fixed effect and week and plot nested within block as the random effects. This model was fitted via the `glmmTMB()` function in the R “glmmTMB” package (Brooks et al. 2017). The effects of treatment on Shannon diversity were evaluated using a linear mixed-effects model with treatment as the fixed effect and week and plot nested within block as the random effects. We checked model assumptions using the simulated scaled residuals generated from the `simulateResiduals()` function in the R “DHARMA” package (Hartig 2024).

Abundance of selected predators and herbivores

To evaluate the effects of pheromone treatment on herbivorous and predator arthropod abundance across taxonomic groups, we constructed generalized linear mixed models using the `lme4` package (Bates et al. 2015). Herbivore and predator datasets were filtered to include taxa that were consistently detected across sampling events. The selected herbivores included *Cicadellidae* (potato leaf hopper), *Galerucinae* (flea beetles), *Miridae* (tarnished plant bug), and *Aphididae* (Aphids); the selected predators included *P. maculiventris*, Neuroptera (lacewings), *Coccinellidae* (lady beetles), *Nabidae* (damselflies), *Araneidae* (spiders), and *Anthrocoridae* (minute pirate bugs). Abundance for each taxonomic group was calculated as the mean number of individuals collected within each combination of block, plot, row, treatment, sampling method, and week.

We modeled herbivore abundance as a function of treatment, taxonomic group, and their interaction using a negative binomial GLMM. The full model included the interaction between treatment and taxonomic group as fixed effects, with plot nested within block and week included as random effects. A negative binomial error distribution was used to account for overdispersion

in count data. Post hoc comparisons among treatment levels within each taxonomic group were performed using pairwise contrasts of estimated marginal means (EMMs) with the emmeans package (Lenth, 2024).

Egg predation in the field

To compare the egg predation rates between the control and the pheromone-treated plots, we fit a generalized linear model with the number of predated eggs as the response, and plot treatment type and week as the predictors. We used a negative binomial distribution with a log link function to account for data overdispersion. The model was fitted via the glmmTMB() function in the R “glmmTMB” package (Brooks et al. 2017). We checked model assumptions using the simulated scaled residuals generated from the simulateResiduals() function in the R “DHARMA” package (Hartig 2024). We used the Wald chi-square test to assess predictor significance using the Anova() function (type II sums of squares) in the R “car” package (Fox & Weisberg 2018). The analysis was performed in R version 4.3.3 (R Core Team 2024).

Egg predation in the greenhouse

We analyzed the probability of egg consumption by four predator species (*P. maculiventris*, *C. maculata*, *L. grandis*, and *C. rufilabris*) using a generalized linear mixed model fitted with the glmmTMB package (Brooks et al. 2017). The response variable was the number of eggs consumed relative to the number available at the start of the assay, modeled as a binomial response. Fixed effects included treatment, predator life stage/species, their interaction, and assay length as a covariate. The species *C. maculata* was separated to analyze the treatment effect on egg consumption in larvae and adult stages. *Coleomegilla maculata* was selected because it represented the highest proportion of lady beetles observed in the field, while *L. grandis* was included due to

its role as an important specialist predator of the Colorado potato beetles. All analysis was conducted in R version 4.4.0 (R Core Team 2025).

Results

Herbivore abundance and community structure

While the abundance of the target herbivore, the Colorado potato beetle, was reduced by the pheromone, none of the other herbivores were affected. On average, the number of Colorado potato beetle larvae was significantly lower in the pheromone-treated plots compared to the control plots ($Z = -2.51$; $p = 0.01$; Fig. 1).

The presence of the synthetic predator pheromone did not significantly alter herbivore abundance, richness, or diversity of the rest of the herbivore community over the seven-week sampling period (Appendix S1; Fig. S1; Table S1). The herbivore community was dominated by potato leafhoppers, which comprised approximately 79% of all herbivores observed, followed by the Colorado potato beetles (6%), aphids (5%), tarnished plant bugs (5%), and flea beetles (4%). Together, these taxa accounted for approximately 99% of the herbivore community.

Aphids, potato leafhoppers, flea beetles, and tarnished plant bugs did not differ significantly in abundance between the control and pheromone treatments (Fig. 2; Table 1).

Predator community structure

Exposure to the synthetic predator pheromone did not significantly alter predator abundance, richness, or diversity over the seven-week sampling period (Appendix S1; Fig. S2; Table S2).

The predator community was dominated by lady beetles, which comprised approximately 48% of all predators observed, followed by minute pirate bugs (22%), spiders (12%), damsel bugs

(6%), spined soldier bugs (5%), and lacewings (4%). Together, these taxa accounted for approximately 97% of the predator community.

The most abundant predators in the potato field—minute pirate bugs, spiders, lady beetles, damsel bugs, lacewings, and spined soldier bugs, did not differ significantly in abundance between the control and pheromone treatments (Fig. 3; Table 2).

Predation rates of Colorado potato beetle eggs in the field

Overall egg predation rates over a two-month period were higher in the pheromone-treated plots compared to the control plots ($\chi^2 = 11.7$; $df = 1$; $p < 0.001$; Fig. 4).

Predation of Colorado potato beetle eggs in greenhouse assays

There is no significant effect of pheromone treatment on the proportion of eggs consumed by adult *C. maculata* ($Z = 0.09$; $p = 0.92$; Fig. 5), *C. maculata* larvae ($Z = 0.83$; $p = 0.41$); *C. rufilabris* ($Z = 0.52$; $p = 0.60$; Fig. 5), or *L. grandis* ($Z = -1.30$; $p = 0.19$; Fig. 5).

There is a significant effect of pheromone treatment on the proportion of eggs consumed by *P. maculiventris* ($Z = -3.69$; $p < 0.001$; Fig. 5) where stink bug adults consumed higher proportions of eggs when the pheromone was present.

Discussion

While the abundance of the target herbivore (the Colorado potato beetle) was reduced by exposure to the synthetic predator pheromone, the abundance, species richness, and diversity of herbivores and natural enemies in the field were not altered. Despite these neutral community responses, predation on beetle eggs was significantly higher in pheromone-treated plots, suggesting that the cue elicited changes in predator behavior rather than changes in predator abundance. Greenhouse assays further revealed that *P. maculiventris* increased its feeding rate in

the presence of the pheromone, demonstrating that the compound can enhance consumptive effects for at least one key predator species. In contrast, *C. maculata* lady beetle larvae and adults, lacewing larvae, and *L. grandis* showed no detectable response. Collectively, although predators varied in their feeding responses to the synthetic pheromone, the net outcome positively favored future use of the pheromone in biological control of the Colorado potato beetle.

The lack of community-wide responses by herbivores may reflect differences in how taxa perceive or respond to predator-associated volatiles. Most herbivores in this system, aside from the Colorado potato beetle, are not known to be preyed upon by *P. maculiventris* and therefore may have little incentive to respond directly to its cues. Although previous work demonstrates that predator pheromones can alter plant physiology and quality, both in potatoes exposed to the *P. maculiventris* pheromone (Aflitto & Thaler, 2020; Martinez et al., 2025) and in other systems where plants respond to predator-derived cues (Helms et al., 2017), the herbivores in our study did not show detectable changes in a way that suggested response to plant quality. Consistent with this, the Colorado potato beetle responded to predator-pheromone-induced changes in plant quality in earlier experiments, but we found no evidence that other herbivores reacted similarly in our field surveys. Many chewing herbivores adjust behavior rather than population numbers in response to predator cues (Kaplan et al., 2014), though such subtle shifts may have gone undetected as we did not measure behavior. Additionally, herbivores such as leafhoppers and aphids rely heavily on plant nutritional quality and phloem chemistry, traits that may not be strongly influenced by exposure to the predator pheromone (Hewer et al., 2010). Chewing herbivores like flea beetles, despite sharing a feeding mode with Colorado potato beetle larvae,

may also differ in how they perceive and respond to predator-associated volatiles (Li et al., 2024).

Aggregation pheromones have been shown to attract adult *P. maculiventris* in field settings, yet such recruitment was not detected in our experiment (Kelly et. al 2014). Predators may vary widely in their responses to the pheromone. Notably, *C. maculata* has been shown to respond to a component of the *P. maculiventris* pheromone under field conditions (Zhu et al. 1999; Kaplan 2012), suggesting capacity for attraction in at least some members of the predator guild. There is evidence that predators can be attracted by HIPV's in field settings; for example, methyl salicylate reliably pulls in diverse predator guilds, showing that single semiochemicals can serve as attractive cues (James, 2005; Zhang et al., 2006; Zhu & Park, 2005). However, predator abundance in our fields remained unchanged across treatments, and predators did not appear to increase their presence in plots with higher Colorado potato beetle larval abundance. This indicates that the pheromone was neither a deterrent nor an attractive cue at the plot scale, nor were any patches of larvae serving as a prey hotspot. However, predation rates on Colorado potato beetle eggs changed in the field, prompting us to examine predator feeding behavior in greenhouse assays, where we did observe species-specific responses to the pheromone. *Podisus maculiventris* increased feeding, which may be due to the species' established use of aggregation pheromones in mate attraction and prey sharing (Sant'Ana et al., 1997). Alternatively, increased egg consumption may reflect a behavioral response to aggregation cues that elevate local predator density, as higher conspecific densities commonly alter per-capita feeding rates through interference or competitive interactions in biological control systems (DeLong & Vasseur, 2011). Although not statistically significant, *L. grandis* exhibited a moderate increase in egg consumption under pheromone exposure, suggesting potential responsiveness among some

beetle predators. Not all predators exhibited behavioral responses to the pheromone treatment. *C. maculata* adults did not respond to the predator pheromone. While there is evidence of adult response to an individual component of the pheromone (Zhu et al. 1999; Kaplan 2012) this sensitivity may not translate into attraction or altered feeding behavior in response to a more complex pheromone blend. Similarly, *C. maculata* larvae and lacewings showed no significant differences in feeding rates, suggesting that the pheromone does not universally enhance predation across the natural enemy guild. This lack of response is noteworthy given that these larval predators could be at risk of intraguild predation by adult stink bugs (Abram et al., 2014; Mallampalli et al., 2002). Predator-derived cues have been shown to trigger behavioral and physiological responses in insect prey, though whether such effects consistently scale up to influence population abundance or community structure remains poorly resolved (Hermann & Landis, 2017). Such asymmetries highlight the importance of considering functional traits and ecological roles of the larger community when evaluating the potential of semiochemicals as biocontrol tools.

From an applied perspective, our findings demonstrate that predator-derived pheromones can enhance biological control. By increasing predation on Colorado potato beetle eggs, the pheromone has the potential to improve the biocontrol efficiency of key predators, particularly given that Colorado potato beetle feeding damage translates directly into yield loss (Hare, 1990). The low numbers of *P. maculiventris* observed in the field suggest that the increased predation in pheromone-treated plots likely resulted from subtle shifts in predator behavior rather than from increases in predator abundance. This is notable because *P. maculiventris* is difficult to retain in cropping systems and often requires multiple augmentative releases (Aldrich & Cantelo, 1999; Biever & Chauvin, 1992; Tipping et al., 1999); even a small increase in conspecific attraction,

combined with higher per-capita feeding rates, could meaningfully improve its efficacy. Importantly, this enhancement occurred without detectable reductions in the abundance of either beneficial insects or non-target herbivores, indicating that pheromone deployment is unlikely to generate the unintended impacts commonly associated with broad-spectrum insecticides. Collectively, these outcomes support the integration of predator pheromones into push–pull or attract-and-stimulate strategies to strengthen pest suppression in potato systems.

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Figure 1: Number of the Colorado potato beetle larvae in the control and the pheromone-treated plots averaged over a two-month period. Error bars represent standard errors. Asterisk denotes a significant difference between the treatment levels ($p < 0.05$).

Alt text 1: Bar plot showing number of Colorado potato beetle larvae in the control (yellow bar) and the predator pheromone (purple bar) plots. There are less beetles in the plots with the predator pheromone.

Figure 2. Average herbivore number per plot across two months in control and pheromone-treated plots. Herbivores are shown by family and were the major herbivore families found in the field. Aphididae (aphids), Cicadellidae (potato leafhopper), Galerucinae (flea beetles), and Miridae (tarnished plant bugs). Error bars represent standard errors.

Alt text 2: Number of herbivores in each plot (control or pheromone). Listed by different families. From right to left – Aphididae, Cicadellidae, Galerucinae, Miridae.

Figure 3. Average predator number per plot across two months in control and pheromone treated plots. Predators are shown by family and are the most abundant in the field, Anthocoridae, Araneidae, Coccinellidae, Nabidae, and Neuroptera. Error bars represent standard errors.

Alt text 3: Number of predators in each plot (control or pheromone). Listed by different families. From right to left – Anthocoridae, Araneidae, Chrysopidae/Hemerobiidae, Coccinellidae, Nabidae, Pentatomidae.

Figure 4. Overall beetle egg predation rates in the control and the pheromone-treated plots over a two-month period. Error bars represent standard errors. Asterisk denotes a significant difference between the treatment levels ($p < 0.05$).

Alt text 4: Predation rates of predators found in potato field. Rates represent eggs consumed every three days in control or pheromone treated plots.

Figure 5. Percent of the Colorado potato beetle eggs consumed by five predator species/life stages when exposed to predator pheromone or control volatiles in greenhouse assays. Panels from left to right show *Coleomegilla maculata* adults, *Chrysoperla rufilabris*, *Lebia grandis*, *Podisus maculiventris*, and *Coleomegilla maculata* larvae. Error bars represent standard errors. Asterisks denote significant differences between treatment levels ($p < 0.05$).

Alt text 5: Percent of Colorado potato beetle eggs consumed by predators in greenhouse assays. From right to left – *Coleomegilla maculata* adults, *Coleomegilla maculata* larvae, *Lebia grandis*, *Chrysoperla rufilabris*, *Podisus maculiventris*.

Taxonomic group	Control / Pheromone ratio	SE	Z	p-value
<i>Aphididae</i>	1.10	0.09	1.09	0.28
<i>Cicadellidae</i>	1.07	0.07	1.04	0.29
<i>Galerucinae</i>	0.93	0.09	-0.72	0.47
<i>Miridae</i>	1.10	0.09	0.14	0.89

Table 1. Control /pheromone abundance ratios ($\pm$ SE) for dominant herbivore taxa. Ratios were estimated from generalized linear mixed models; values significantly different from 1 would indicate treatment effects. No significant differences were detected ($p < 0.05$).

Taxonomic Group	Control / Pheromone ratio	SE	Z	p-value
<i>Anthocoridae</i>	1.11	0.17	0.69	0.49
<i>Araneidae</i>	0.81	0.25	-0.68	0.50
<i>Coccinellidae</i>	0.99	0.09	-0.14	0.89
<i>Nabidae</i>	0.92	0.24	-0.33	0.74
<i>Neuroptera</i>	1.12	0.35	0.46	0.65
<i>Podisus</i>	1.11	0.38	0.36	0.76

Table 2. Control/pheromone abundance ratios ($\pm$ SE) for dominant predator taxa. Ratios were estimated from generalized linear mixed models; values significantly different from 1 would indicate treatment effects. No significant differences were detected ($p < 0.05$).

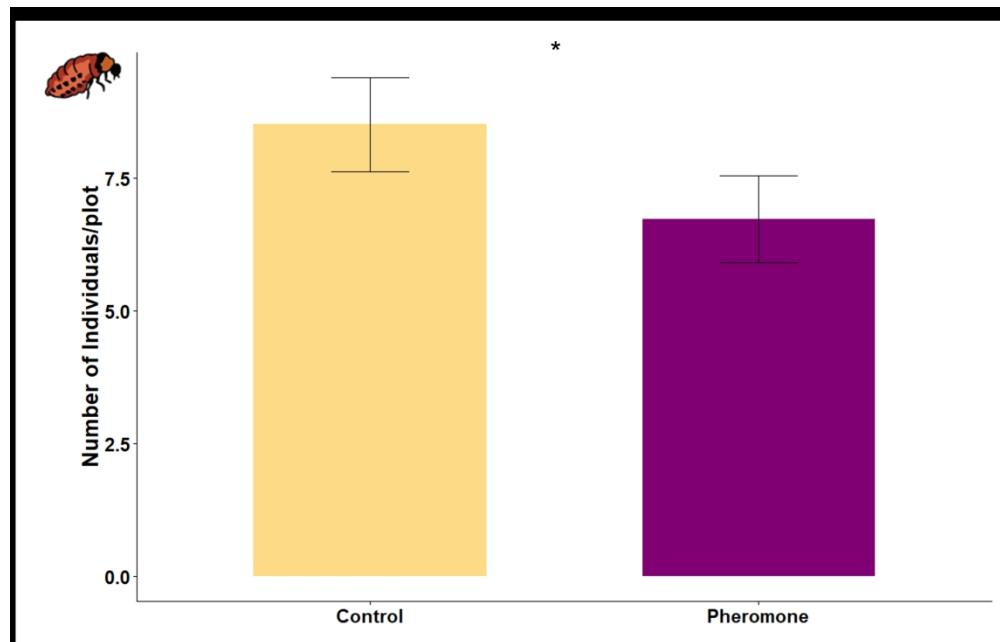


Figure 1: Number of the Colorado potato beetle larvae in the control and the pheromone-treated plots averaged over a two-month period. Error bars represent standard errors. Asterisk denotes a significant difference between the treatment levels ($p < 0.05$).

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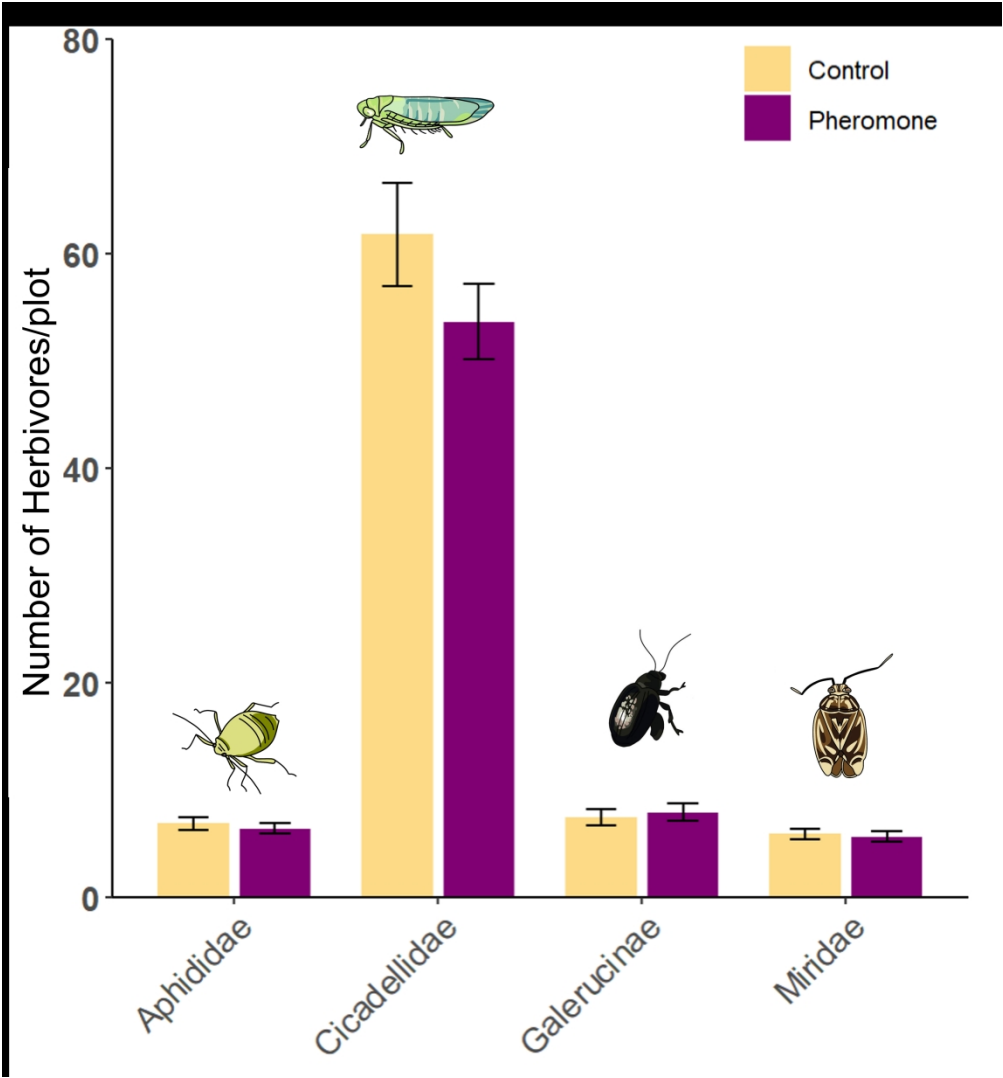


Figure 2. Average herbivore number per plot across two months in control and pheromone-treated plots. Herbivores are shown by family and were the major herbivore families found in the field. Aphididae (aphids), Cicadellidae (potato leafhopper), Galerucinae (flea beetles), and Miridae (tarnished plant bugs). Error bars represent standard errors.

682x733mm (130 x 130 DPI)

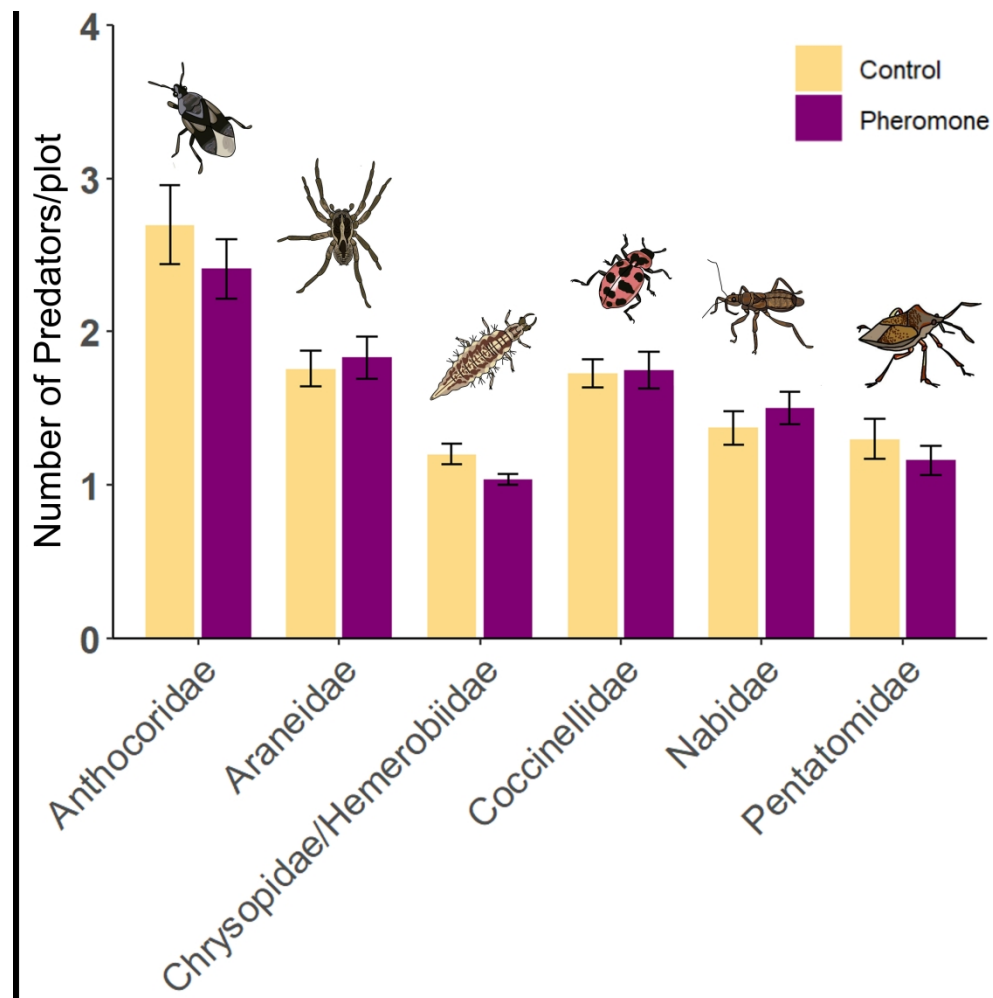


Figure 3. Average predator number per plot across two months in control and pheromone treated plots. Predators are shown by family and are the most abundant in the field, Anthocoridae, Araneidae, Coccinellidae, Nabidae, and Neuroptera. Error bars represent standard errors.

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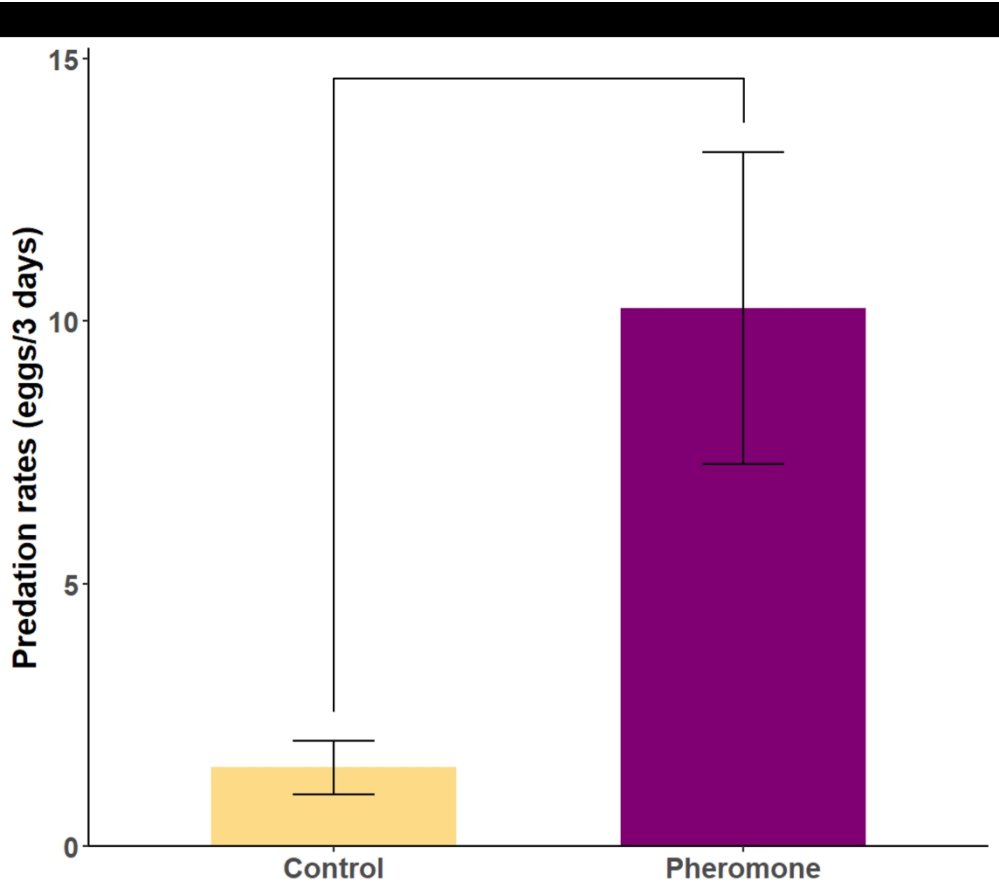


Figure 4. Overall beetle egg predation rates in the control and the pheromone-treated plots over a two-month period. Error bars represent standard errors. Asterisk denotes a significant difference between the treatment levels ($p < 0.05$).

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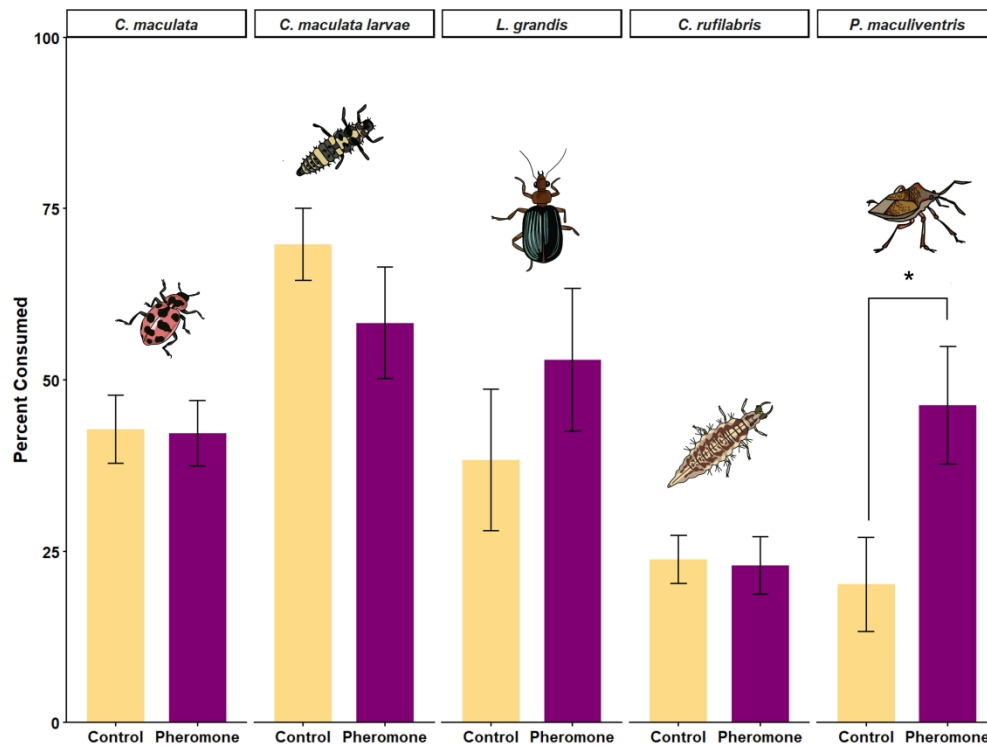
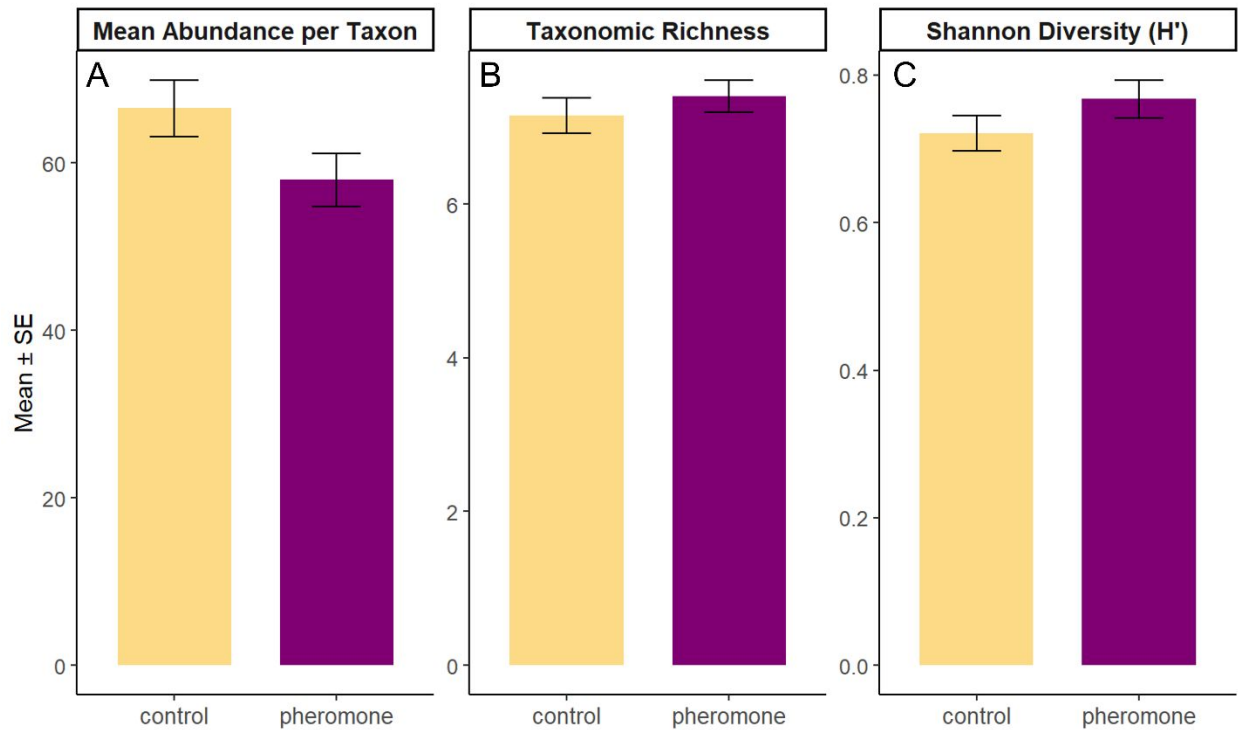


Figure 5. Percent of the Colorado potato beetle eggs consumed by five predator species/life stages when exposed to predator pheromone or control volatiles in greenhouse assays. Panels from left to right show *Coleomegilla maculata* adults, *Chrysoperla rufilabris*, *Lebia grandis*, *Podisus maculiventris*, and *Coleomegilla maculata* larvae. Error bars represent standard errors. Asterisks denote significant differences between treatment levels ($p < 0.05$).

967x725mm (130 x 130 DPI)

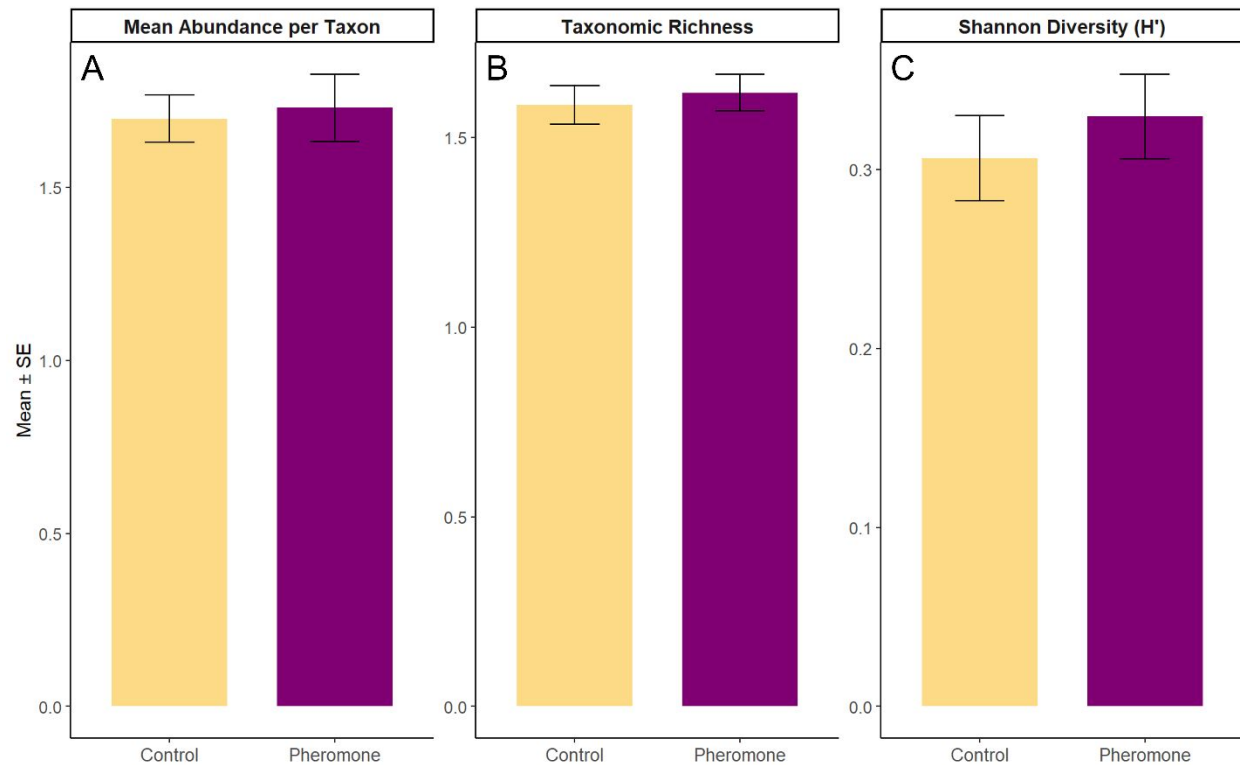
Supplemental/extra figures:



Supplemental Figure S1. A) Mean abundance, B) Richness, C) Diversity of herbivore taxa in the control and the pheromone-treated plots over a two-month period. Error bars represent standard errors.

Table S1. Statistical results examining the effects of pheromone treatment on herbivore abundance, richness, and diversity.

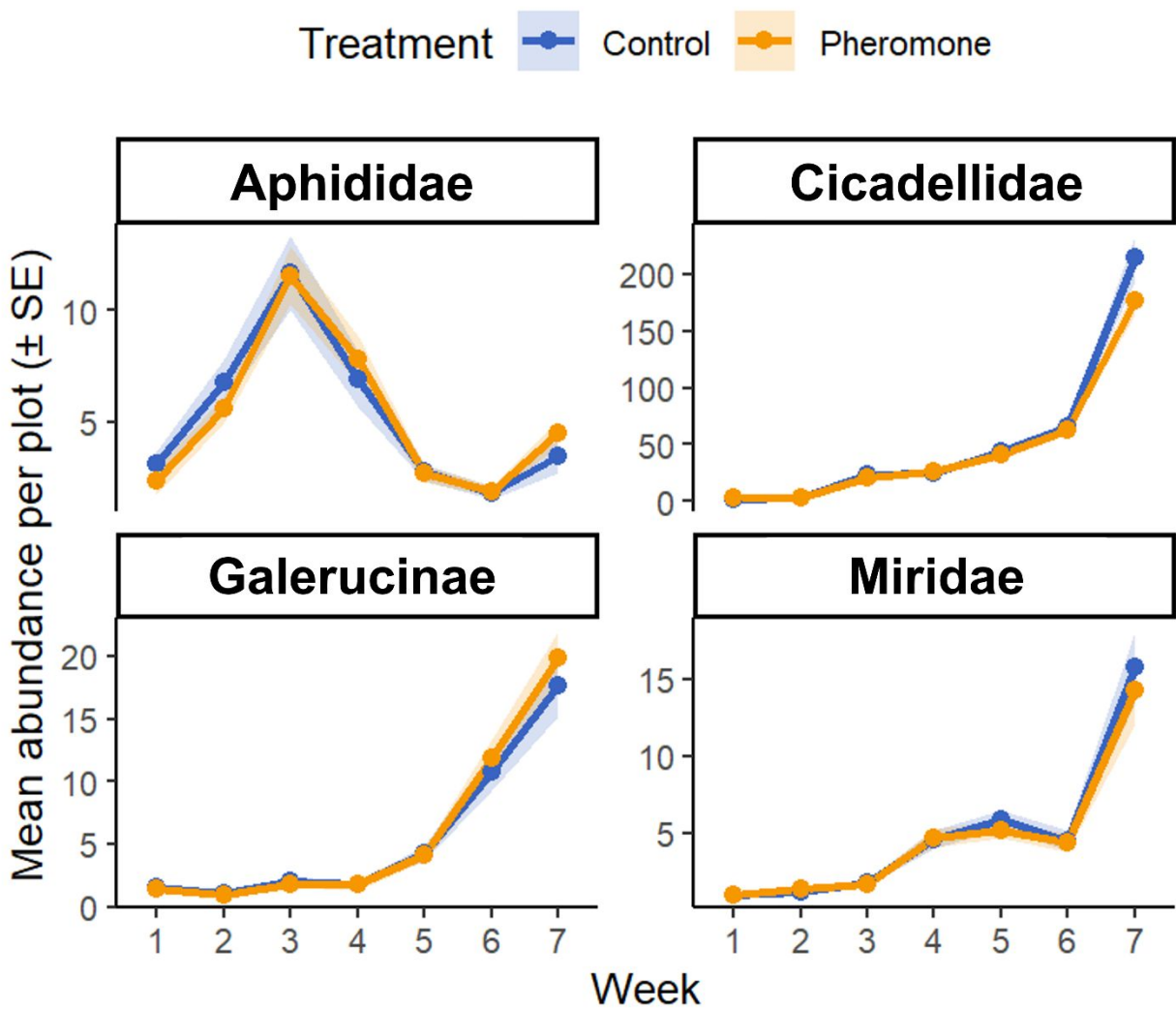
Response variable	Test statistic	df	<i>p</i> -value
<i>Abundance</i>	$t = -0.91$	139.5	0.36
<i>Richness</i>	$Z = -0.22$	---	0.83
<i>Diversity</i>	$t = 0.36$	139.9	0.72



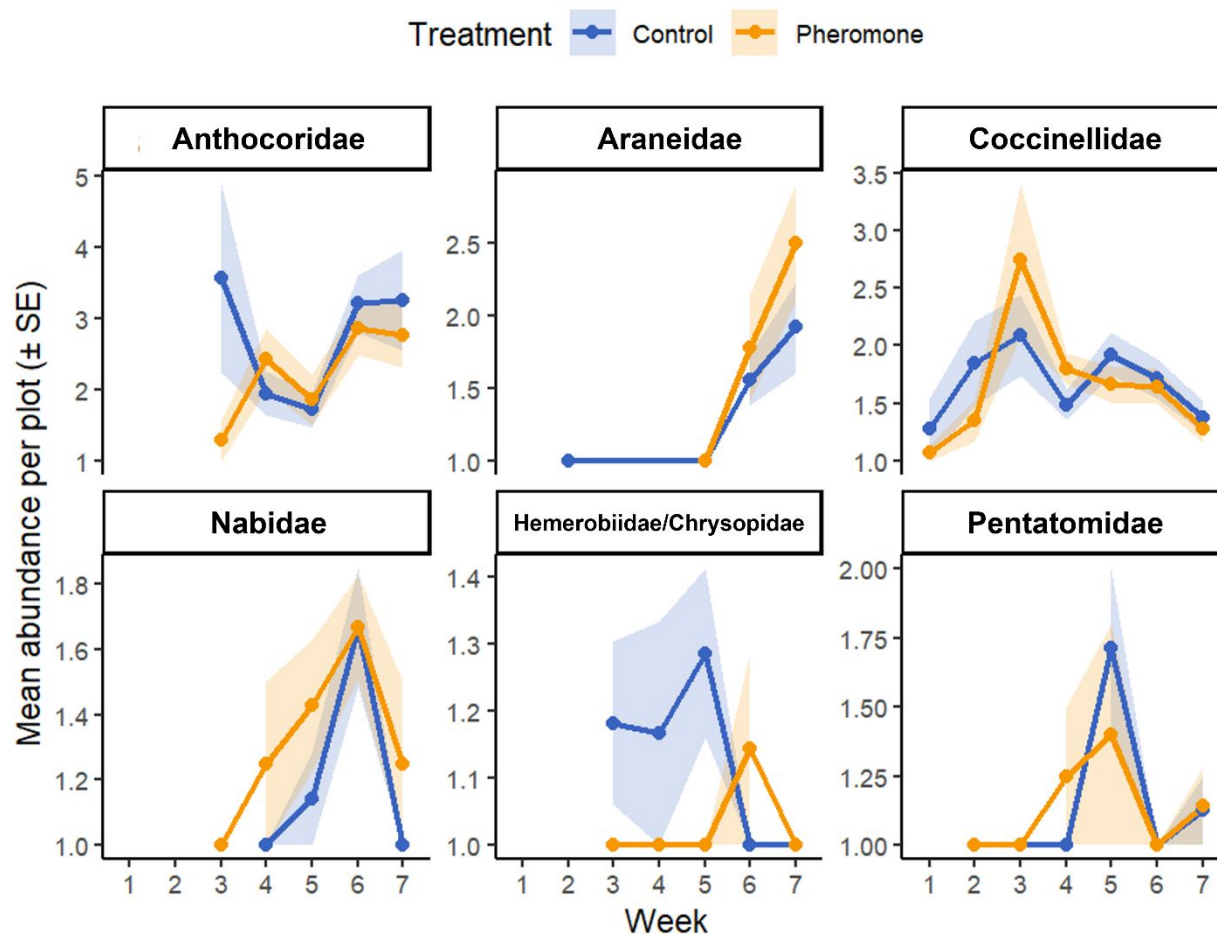
Supplemental Figure S2. A) Mean abundance, B) Richness, C) Diversity of predator taxa in the control and the pheromone-treated plots over a two-month period. Error bars represent standard errors.

Table S2. Statistical results examining the effects of pheromone treatment on predator abundance, richness, and diversity.

Response variable	Test statistic	df	<i>p</i> -value
<i>Abundance</i>	$t = -0.12$	124	0.90
<i>Richness</i>	$Z = 0.51$	---	0.61
<i>Diversity</i>	$t = 0.84$	134	0.40



Supplemental Figure S3. Pest abundance shown over the seven-week period. Each line represents the average abundance of herbivores per plot over time. The blue line represents control treatment, and the orange line represents the pheromone treatment.



Supplemental Figure S4. Predator abundance shown over the seven-week period. Each line represents the average abundance of predators per plot over time. The blue line represents control treatment, and the orange line represents the pheromone treatment.